

Element G: Construction of a Testable Prototype

Prototype Construction of Design #1 - Duct Tape Belt Pouch.

Date: 5/6/24

Tools and Manufacturing Methods used

Step 1: Get some duct tape and cut a small square portion

Step 2: Make a small duct tape pocket that wraps around then belt that can fit the tracker

Step 3: Lastly, attach the duct tape pocket to the belt



Materials name: Duct Tape, Tracker, Belt

Description: The tape is used to make a pocket for the belt, the tracker is the item being used for our project to track the person, and the belt is a place holder to hold the tracker and keep it hidden.

Where material was purchased: All of our materials were purchased from Walmart

Cost per unit # units: The duct tape was around \$3.00, the tracker was \$20.00, and the belt was \$12.00

Total cost of items: \$35.00

How will this prototype allow you to collect objective, measurable data?:

- This item tracks and holds the tracker in a hidden pocket which fits our objects. It also tracks the person so you get an accurate placement of where that person is.

Prototype Construction of Design #2: **Duct Tape Watch Pocket**
Date: **5/8/2024**

Tools and Manufacturing Methods used

Step 1: Get Duct tape and once again, make a pocket for for tracker

Step 2: Attach the pocket to the watch



Materials name: Duct Tape, Watch, Tracker

Description: The duct tape is used to make a pocket for the tracker and attach it to the watch. The watch is the item we are using to hold the tracker, the tracker is the main part of the project so you can track the person.

Where material was purchased: All the materials were purchased from Walmart

Cost per unit # units: The duct tape was around \$3.00, the tracker was \$20.00, and the watch was \$7.00

Total cost of items: \$30.00

How will this prototype allow you to collect objective, measurable data?:

- This tracker allows us to get a measurement of how far away the person is and where they are at. It doesn't keep the tracker as well hidden as the other prototype does.

We choose to go with "Design #1, the Duct Tape Belt Pouch". This was because it is less obvious, more accessible since more people wear belts than watches, and most watches today already have trackers, most belt's do not.